

Md Joshem Uddin

Postdoctoral Research & Teaching Associate, University of Georgia

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 🌐 [Github](#) | 🌐 [Personal Website](#) | [G](#) [Google Scholar](#)

Research Interests

Mathematical and computational methods: Topological data analysis; persistent homology and multiparameter persistence; computational topology; mathematical modeling and scientific computing.

Machine learning and data science: Topological and geometric machine learning; graph representation learning; graph neural networks and transformers; temporal and spatio-temporal learning.

Applications: Biomolecular data analysis and molecular property prediction; biomedical image analysis; cyberattack and anomaly detection in energy systems; power-grid monitoring and resilience.

Education

PhD in Mathematics **2026**

The University of Texas at Dallas, Richardson, TX

Dissertation: *Topology-Aware Graph Learning: From Static Representations to Temporal Networks*

Advisor: [Prof. Baris Coskun](#).

MS in Mathematics **2025**

The University of Texas at Dallas, Richardson, TX

(Applied Mathematics Concentration)

CGPA: 3.804/4.00

Graduate Certificate in Data Science **2025**

The University of Texas at Dallas, Richardson, TX

(12 graduate credit hours)

MS in Applied Mathematics **2018**

University of Dhaka, Dhaka, Bangladesh

GPA: 4.00/4.00

BS in Mathematics **2016**

University of Dhaka, Dhaka, Bangladesh

CGPA: 3.95/4.00

Academic Appointments

Postdoctoral Research & Teaching Associate Aug. 2026–Present

Department of Mathematics, University of Georgia

Athens, GA

Graduate Research and Teaching Assistant Aug. 2021–Aug. 2026

Department of Mathematical Sciences, The University of Texas at Dallas

Richardson, TX

Lecturer Nov. 2018–Aug. 2021

Department of Applied Mathematics, University of Dhaka

Dhaka, Bangladesh

Lecturer May 2018–Nov. 2018

Department of Mathematics, American International University-Bangladesh

Dhaka, Bangladesh

Publications

Published**2026**

- [1] *From Nodes to Pixels: Topological and Structural Two-View Graph Imaging*. M. J. Uddin, S. Changani, S. K. Navuluru, C. Akcora, and B. Coskunuzer. **NeurIPS**, 2026.
- [2] *TopoFormer: Topology Meets Attention for Graph Learning*. M. J. Uddin, A. Tola, C. Akcora, and B. Coskunuzer. **International Conference on Learning Representations (ICLR)**, 2026.
- [3] *T3Former: Temporal Graph Classification with Topological Machine Learning*. M. J. Uddin, S. Changani, and B. Coskunuzer. **AAAI Conference on Artificial Intelligence (AAAI)**, 2026.
- [4] *GraphTCL: Topology-Aware Contrastive Learning for Graph Classification*. A. S. Korkmaz, S. Coskunuzer, and M. J. Uddin. **NeuroComputing**, 2026. *Conducted with two undergraduate students under my supervision.*
- [5] *MP-Grid: Detecting Power Grid Outages with Topological Machine Learning*. M. J. Uddin, D. Olojede, R. A. Jacob, J. Zhang, and B. Coskunuzer. **Applied Energy**, 2026.
- [6] *PowerBench: A Benchmark Suite for Monitoring and Security in Distribution Systems*. M. J. Uddin, R. A. Jacob, D. Olojede, J. Zhang, and B. Coskunuzer. **Journal of Data-centric Machine Learning Research (DMLR)**, 2026.

2025

- [7] *SCNode: Spatial and Contextual Coordinates for Graph Representation Learning*. M. J. Uddin, A. Tola, V. Sikand, C. Akcora, and B. Coskunuzer. **Transactions on Machine Learning Research (TMLR)**, 2025.
- [8] *Topology Informed Transformer for Cyber Attack Detection in Grid-Connected PV Systems*. M. J. Uddin, D. Olojede, R. A. Jacob, B. Coskunuzer, and J. Zhang. **IEEE Transactions on Sustainable Energy**, 2025.
- [9] *Cyber Attack Detection in Electric Vehicle Charging Stations Using Topological Data Aided Learning*. M. J. Uddin, R. A. Jacob, J. Wang, B. Coskunuzer, and J. Zhang. **IEEE Power & Energy Society General Meeting (PESGM)**, 2025.
- [10] *OcuViT: A Vision Transformer-Based Approach for Automated Diabetic Retinopathy and AMD Classification*. F. Ahmed and M. J. Uddin. **Journal of Imaging Informatics in Medicine**, 2025.

2024

- [11] *Cyber Attack Detection in Distribution Networks with Topological Data Analytics Aided Learning*. M. J. Uddin, D. Olojede, R. A. Jacob, B. Coskunuzer, and J. Zhang. **IEEE Kansas Power and Energy Conference (KPEC)**, 2024.

2022

- [12] *Estimating Option Prices with Discrete Dividend Payment Using Finite Difference Method and Monte Carlo Simulation: A Comparative Study*. M. J. Uddin, S. Hossain, A. Hossain, S. Parvin, A. Rahman, and M. H. Sohel. **Journal of Applied Mathematics and Computation**, 6(4), 472–481, 2022.

2021

- [13] *Impact of Horizontal Grid Resolutions for Thunderstorms Simulation over Bangladesh Using WRF-ARW Model*. M. J. Uddin, M. A. Samad, and M. A. K. Mallik. **Dhaka University Journal of Science**, 69(1), 43–51, 2021.
- [14] *Track and Landfall Characteristics of Very Severe Cyclonic Storm Fani over the Bay of Bengal Using*

WRF Model. S. Hossain, M. A. Samad, R. Sultana, M. A. K. Mallik, and M. J. Uddin. **Dhaka University Journal of Science**, 69(2), 101–108, 2021.

Under Review

- [1] *Label Encodings: A Missing Channel for Graph Learning*. **M. J. Uddin**, S. K. Navuluru, S. Kumar, S. Lee, and B. Coskunuzer. Under review at **NeurIPS**, 2026.
- [2] *Node Signal Atlas: Characterizing Node Classification Beyond Homophily*. **M. J. Uddin**, S. Lee, and B. Coskunuzer. Under review at **NeurIPS**, 2026.
- [3] *Temporal Graph Transformers for PMU-Based Cyberattack Detection in Power Grids*. **M. J. Uddin**, D. Olojede, K. Hutchison, R. A. Jacob, B. Coskunuzer, and J. Zhang. Under review at **International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)**, 2026.

Selected Research Projects

Topological Machine Learning for Graph Representation Learning. Developed topology-aware frameworks for graph-level and node-level learning by combining persistent homology, structural descriptors, contextual coordinates, graph neural networks, and transformer architectures.

Temporal Graph Learning with Topological Signals. Designed temporal graph classification models that fuse message-passing representations, topological summaries, spectral descriptors, and attention-based sequence modeling for dynamic networks.

Power Grid Security and Resilience. Built topology-aided learning systems for outage detection, cyberattack detection, EV charging station security, PV system monitoring, and PMU-based power-grid anomaly detection.

Invited Talks and Conference Presentations

Invited Talks

AMS Eastern Sectional Meeting Oct. 3–4, 2026 (Upcoming)

Topology-Informed Transformer for Cyber Attack Detection in Grid-Connected PV Systems

Special Session on Artificial Intelligence and Machine Learning Methods in Topological Data Analysis.
The George Washington University, Washington, D.C.

AMS Southeastern Sectional Meeting Oct. 2025

Topological Transformers for Graph Representation Learning

Special Session on Computational Topology and Geometry in Data Science.
Tulane University, New Orleans, LA.

AMS Southeastern Sectional Meeting Oct. 2025

MP-Grid: Detecting Power Grid Outages with Topological Machine Learning

Special Session on Advances in Applied Topology and Topological Data Analysis.
Tulane University, New Orleans, LA.

SIAM Conference Oct. 2024

WISE-GNN: Enhancing GNNs with Wise Embeddings and Topological Encoding

Conference on Geometric and Topological Approaches in Data Science and Machine Learning.
Baylor University, Waco, TX.

AI Conference Presentations

AAAI Conference on Artificial Intelligence

Jan. 2026

T3Former: Temporal Graph Classification with Topological Machine Learning
Singapore.

[Poster Presentations](#)

ICERM Workshop on Computational Topology

May 2026

TopoFormer: Topology Meets Attention for Graph Learning

Workshop on Computational Topology.

Institute for Computational and Experimental Research in Mathematics, Brown University, Providence, RI.

ATD–AMPS Joint PI Workshop

Oct. 2024

Enhancing Cyber Attack Detection in Grid-Connected PV Systems: A Topological Machine Learning Approach

Algorithms for Threat Detection and Algorithms for Modern Power Systems Joint PI Workshop, Washington, D.C.

TACCSTER

Sept. 2024

WISE-GNN: Enhancing GNNs with Wise Embeddings

Texas Advanced Computing Center, The University of Texas at Austin, Austin, TX.

ATD–AMPS Joint PI Workshop

Oct. 2023

Outage Detection in Power Grids: A Topological Machine Learning Approach

George Mason University, Fairfax, VA.

Teaching Experience**Postdoctoral Research & Teaching Associate**

Aug. 2026–Present

Department of Mathematics, University of Georgia

Athens, GA

- Instructor of record: MATH 2250, Calculus I for Science and Engineering.

Graduate Teaching Assistant

Aug. 2021–Aug. 2026

Department of Mathematical Sciences, The University of Texas at Dallas

Richardson, TX

- Led classroom problem-solving sections for Differential Calculus, Applied Calculus I, Calculus II, Calculus of Several Variables, and Linear Algebra.

Lecturer

Nov. 2018–Aug. 2021

Department of Applied Mathematics, University of Dhaka

Dhaka, Bangladesh

- Courses taught: Coordinate and Vector Geometry; Advanced Linear Algebra; Mathematical Methods; Mathematical Modeling in Biology and Physiology.

Lecturer

May 2018–Nov. 2018

Department of Mathematics, American International University-Bangladesh

Dhaka, Bangladesh

- Courses taught: Engineering Mathematics; Ordinary Differential Equations.

Mentoring

- **UT Dallas, Summer 2025:** Supervised two undergraduate researchers from Texas A&M University and Virginia Tech on topological machine learning for graph classification. Resulting publication: *GraphTCL: Topology-Aware Contrastive Learning for Graph Classification*, **Neurocomputing**, 2026.

- **UT Dallas, Summer 2024:** Supervised one undergraduate researcher from UT San Antonio on molecular property prediction using topological machine learning.
- **University of Dhaka, 2020:** Supervised four senior undergraduate students on numerical methods for option pricing. Resulting publication: *Estimating Option Prices with Discrete Dividend Payment Using Finite Difference Method and Monte Carlo Simulation: A Comparative Study*, **Journal of Applied Mathematics and Computation**, 2022.

Honors and Awards

- **AAAI-26 Scholarship and Volunteer Award**, 2026.
- **First Place**, NSF Algorithms for Modern Power Systems (AMPS) Challenge, 2024.
- **National Science and Technology Fellowship**, Bangladesh, 2016–2017.
- **A. F. Mujibur Rahman Foundation Gold Medal**, MS in Applied Mathematics, University of Dhaka, 2016.
- **A. F. Mujibur Rahman Foundation Gold Medal**, BS in Mathematics, University of Dhaka, 2015.
- **Dean's Award**, Faculty of Sciences, University of Dhaka, 2015.
- **Swargia Anima Rani Saha Memorial Gold Medal**, University of Dhaka, 2015.
- **University Grants Commission Merit Scholarship**, Bangladesh, 2015.

Highlighted Coursework

PURE MATHEMATICS	Real Analysis, Functional Analysis, Complex Analysis, Abstract Algebra, Topology, Combinatorics & Graph Theory
APPLIED MATHEMATICS	Numerical Analysis, Scientific Computing, Optimization, Principles & Techniques in Applied Mathematics, Ordinary Differential Equations
STATISTICS	Probability & Statistics for Data Science, Statistical & Machine Learning
COMPUTER SCIENCE	Algorithms & Data Structures, Machine Learning, Big Data Management, Computer Science I

Technical Skills

Programming	Python, C++, Java, R, MATLAB, Fortran, Mathematica.
Machine Learning	PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, XGBoost, graph neural networks, transformers, sequence models.
Topological Data Analysis	Persistent homology, multiparameter persistence, GUDHI, giotto-tda, Ripser, Dionysus.
Data and Computing	NumPy, pandas, NetworkX, SQL/NoSQL, Apache Spark, MLlib, high-performance and cluster computing.
Scientific Computing	Numerical methods, optimization, simulation, data preprocessing, model evaluation, reproducible experimental pipelines.

Professional Service

- Organizer, Mini-symposium at SIAM Texas-Louisiana Section Conference, 2024.
- Member, Organizing Committee, International Mathematics Conference, University of Dhaka, 2019.

Professional Memberships

- Student Member, American Mathematical Society (AMS).
- Student Member, Association for the Advancement of Artificial Intelligence (AAAI).
- Student Member, Institute of Electrical and Electronics Engineers (IEEE).
- Student Member, Society for Industrial and Applied Mathematics (SIAM).
- Member, Bangladesh Mathematical Society (BMS).

Last updated: June 2026.